

YOUR OFE 2025 REPORT

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2025 Cropping Season

Field G7 | **Research question:** Is corn yield different in areas with legume sod plowed under before first cutting vs. after cutting once?

Welcome to your 2025 farm report. This report summarizes the on-farm nitrogen experiment conducted in Field G7, where plots with legume sod plowed under before first cutting (Treatment) were compared against plots plowed under after cutting once (Control).



Figure 1. Field G7 — treatment layout. ■ Treatment ■ Control

Take-home message: Plowing the sod earlier (Treatment) did produce greener, higher-chlorophyll leaves, a sign of better early-season vigor. Though the in-season plant nitrogen was slightly lower in the treatment, end of season nitrate was slightly higher. Final yield also appeared slightly higher in the unmowed treatment area, relative to the control. Both systems kept the crop adequately supplied with nitrogen, Control CSNT came in just below the optimal threshold, Treatment just above it. Your soil is in good shape overall, and this is a question worth revisiting with another season of data.

Results Summary

Metric	Control	Treatment	Difference	p-value
 NBI (Nitrogen Balance Index)	59.5	59.9	+0.4 (0.7%)	p = 0.888
 Chlorophyll Index	37.4	40.8	+3.4 (9.1%)	p = 0.00239
 Biomass %N (C:N analysis)	3.92%	3.775%	-3.7%	n too small*
 CSNT (Cornstalk Nitrate)	661 ppm	771 ppm	+110 ppm	n = 1
 Corn Yield (bu/ac)	123.8	131.6	+7.8 (6.3%)	p < 0.001

* C:N sample sizes are too small for reliable hypothesis testing (n = 5 Control, n = 4 Treatment). Effect sizes are reported instead.

Corn Yield

The map below shows the **corn yield spatial distribution** across Field G7, displayed in bushels per acre. The blue line marks the field boundary and the red lines separate the treatment strips.

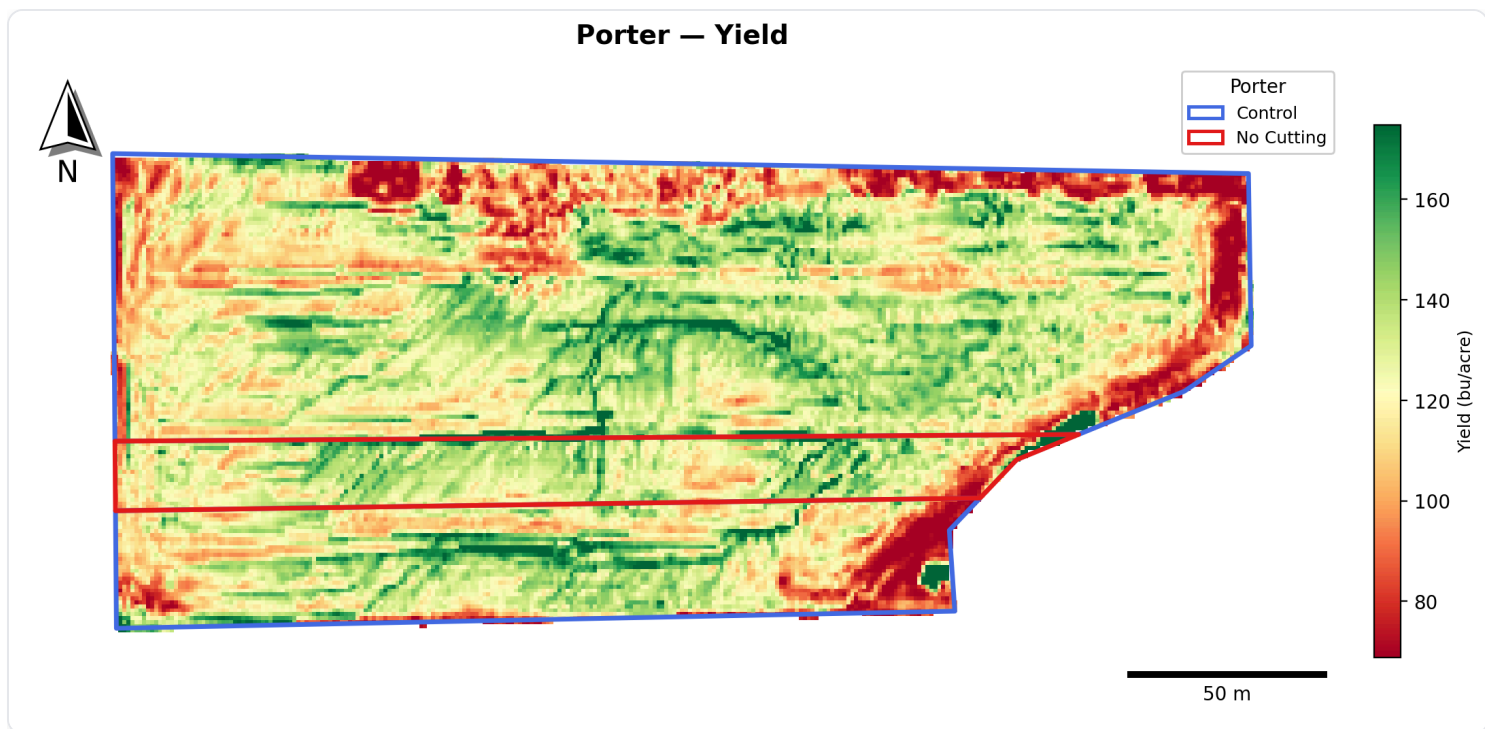


Figure 2. Corn yield map.

[Open Interactive Map](#)

TREATMENT

131.6 ↑ +7.8 vs Control (6.3%)

Control mean: 123.8

EFFECT SIZE

6.3%

relative difference Treatment vs Control

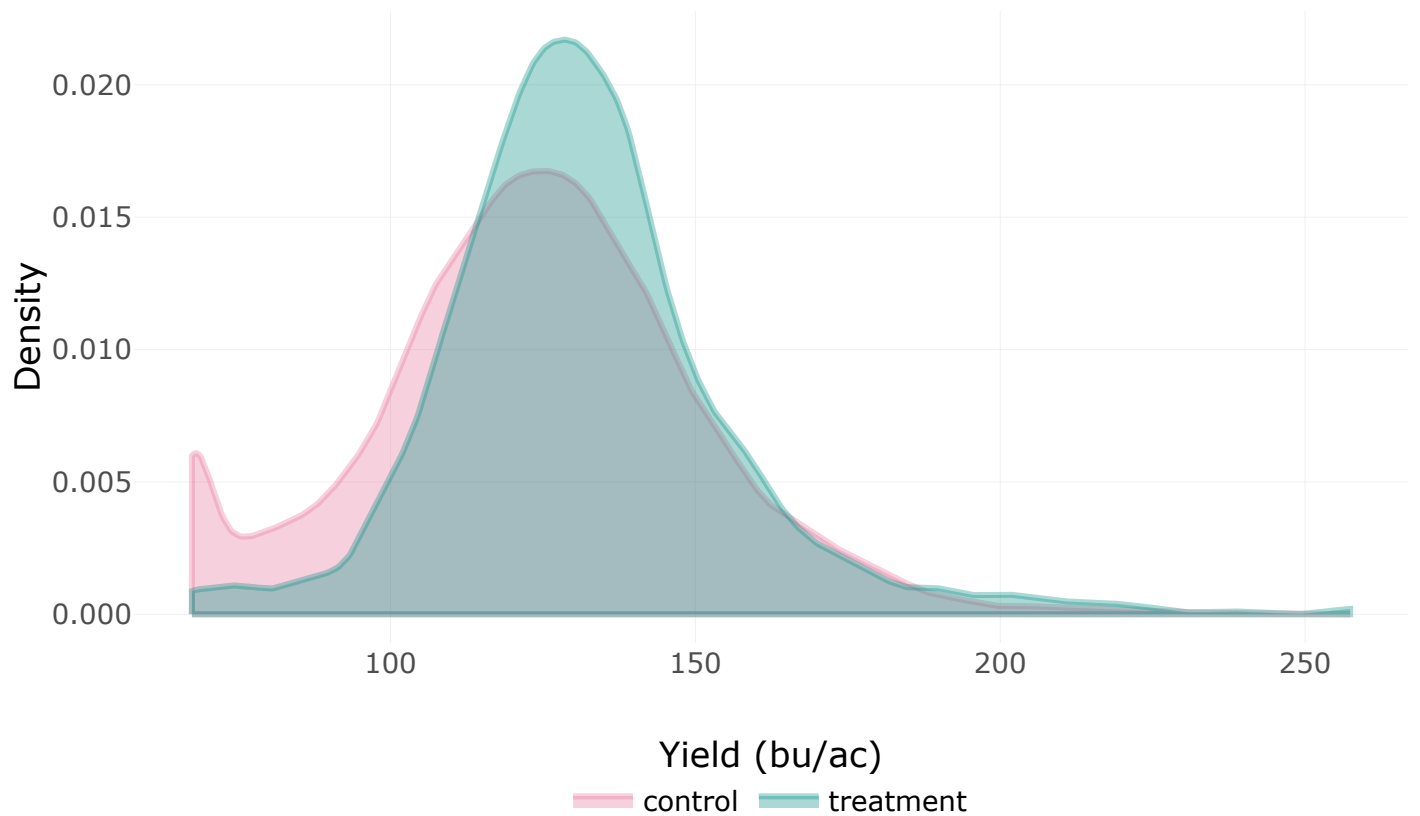
p < 0.001 ✓

Density

Boxplot

Bar Chart (Mean ± SE)

Corn Yield



n = 19948 Control pixels · n = 2981 Treatment pixels. Two-sample Welch t-test; large n makes p-values very sensitive — interpret effect size (% difference) alongside significance.

Nitrogen Balance Index (NBI)

The **Nitrogen Balance Index (NBI)** is measured with a handheld Dualex sensor and gives us a real-time snapshot of how well the plant is supplied with nitrogen during the season. It works by comparing how much chlorophyll the leaf has built relative to stress compounds (flavonoids), so a higher NBI means the plant is using nitrogen efficiently and not showing signs of stress. We took two leaf readings per plant and averaged them, then compared Treatment vs Control strips with a statistical test (two-sample t-test) to see if there was a real difference.

TREATMENT

59.9 ↑ +0.4 vs Control (0.7%)

Control mean: 59.5

EFFECT SIZE

0.7%

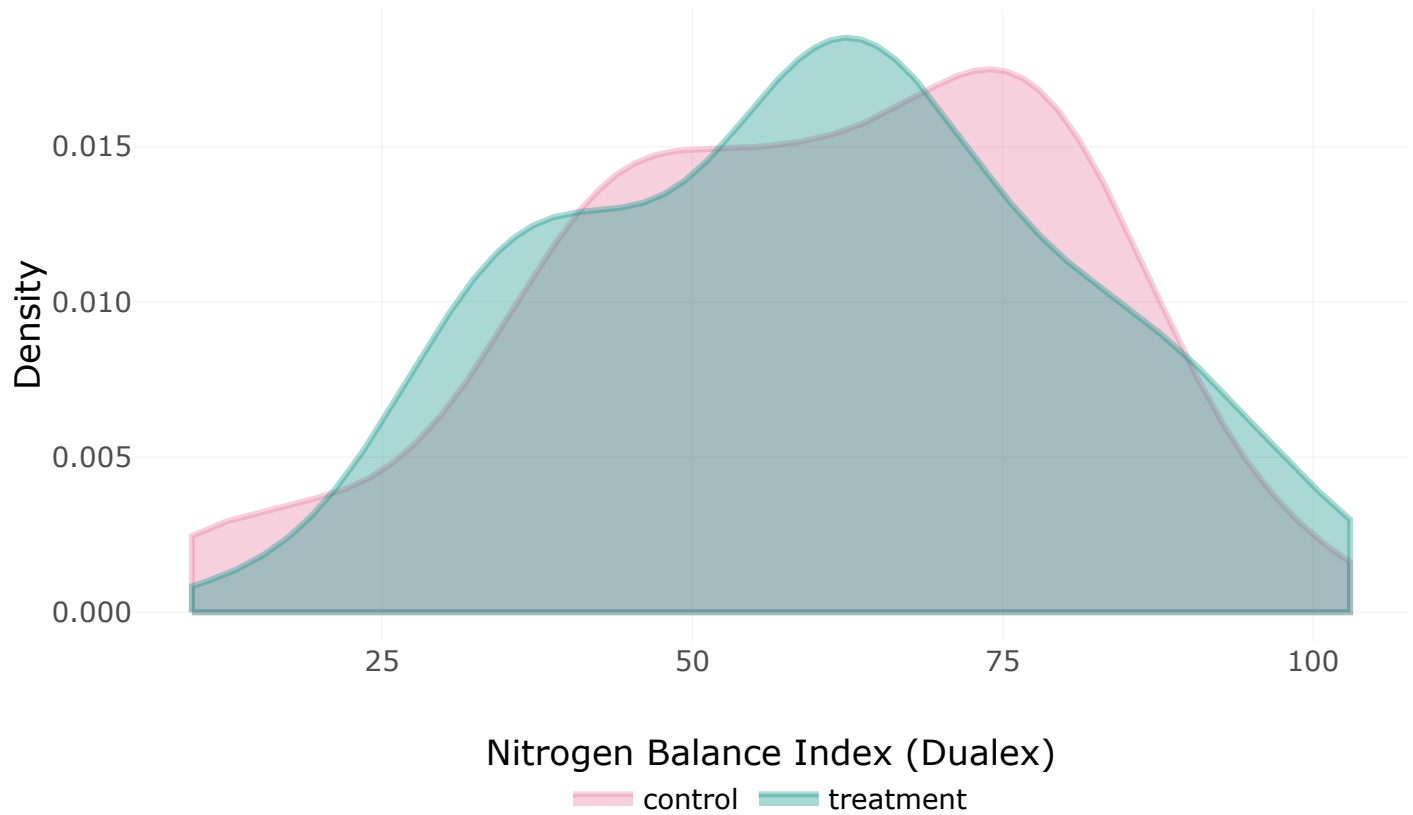
relative difference Treatment vs Control

p = 0.888 ✓

Density

Boxplot

Nitrogen Balance Index (NBI) — Plant Average



Chlorophyll (Chl)

The **Chlorophyll Index (Chl)** is also measured with the Dualect sensor and estimates the amount of chlorophyll in the leaf. Chlorophyll is directly linked to the plant's capacity for photosynthesis and is a reliable proxy for nitrogen sufficiency. Higher values indicate greener, more nitrogen-sufficient leaves. As with NBI, two leaves per plant were averaged and treatments compared with a t-test.

TREATMENT

40.8 ↑ +3.4 vs Control (9.1%)

Control mean: 37.4

EFFECT SIZE

9.1%

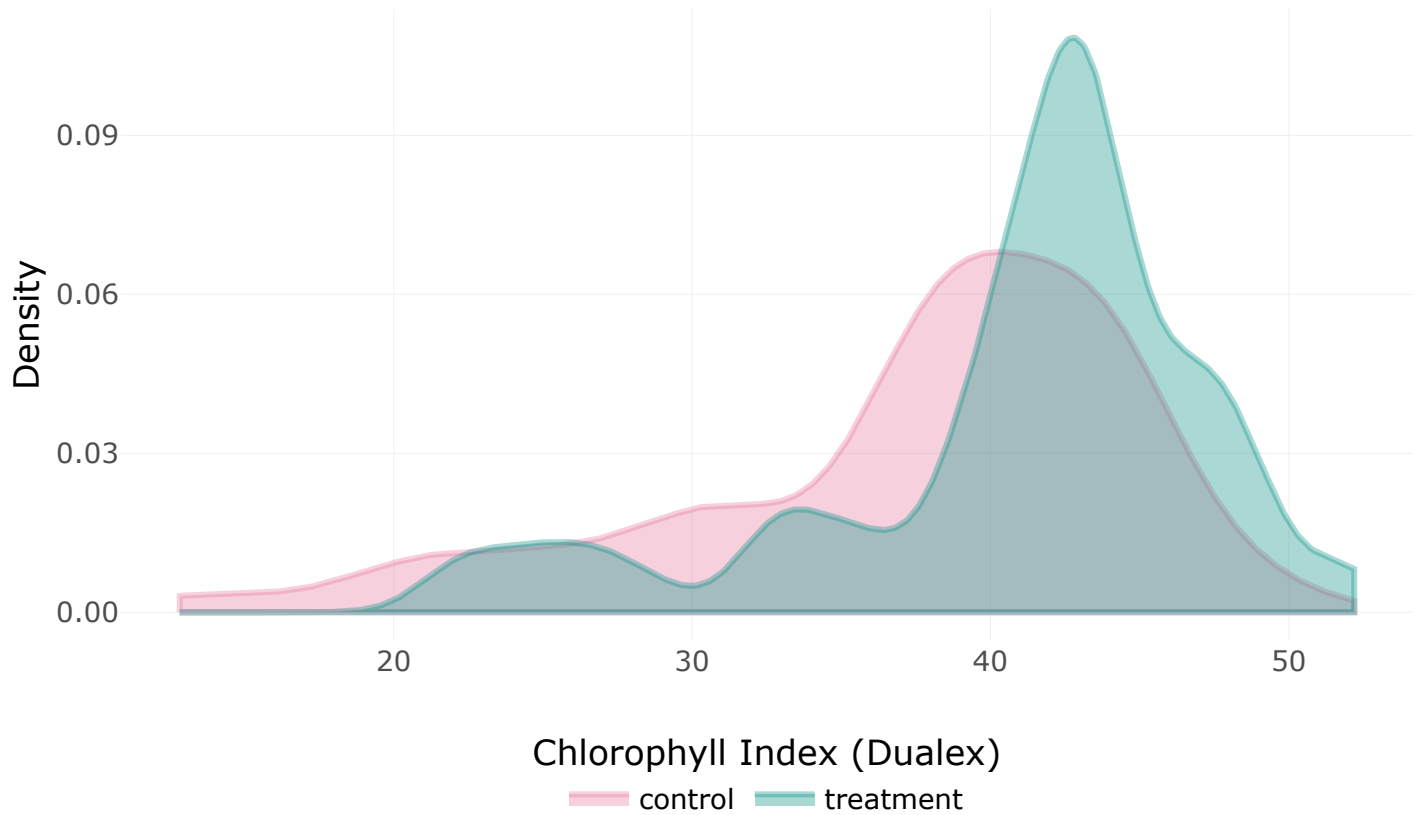
relative difference Treatment vs Control

p = 0.00239 ✓

Density

Boxplot

Chlorophyll (Chl) — Plant Average



C:N in Corn Biomass

We collected corn plant samples from both Treatment and Control strips and sent them to the lab for nitrogen analysis. This tells us exactly how much nitrogen ended up inside the plant — not just what was in the soil or available in theory, but what the crop actually took up. Because the number of samples is small, we can't run a standard statistical test, but we report the effect size (Cohen's d) to give you a sense of how meaningful the difference is in practical terms.

Total Nitrogen in Biomass (%N)

This measures what percentage of the plant tissue is nitrogen. A higher %N means the plant had good access to nitrogen at the time of sampling. In this case, the Control actually came in slightly higher than the Treatment, a small but noteworthy reversal compared to what the chlorophyll data suggested. It's a reminder that mid-season leaf readings and actual tissue nitrogen don't always tell the same story.

TREATMENT — MEAN %N

3.775 ↓ -3.7% vs Control

Control mean: 3.92

COHEN'S D (EFFECT SIZE)

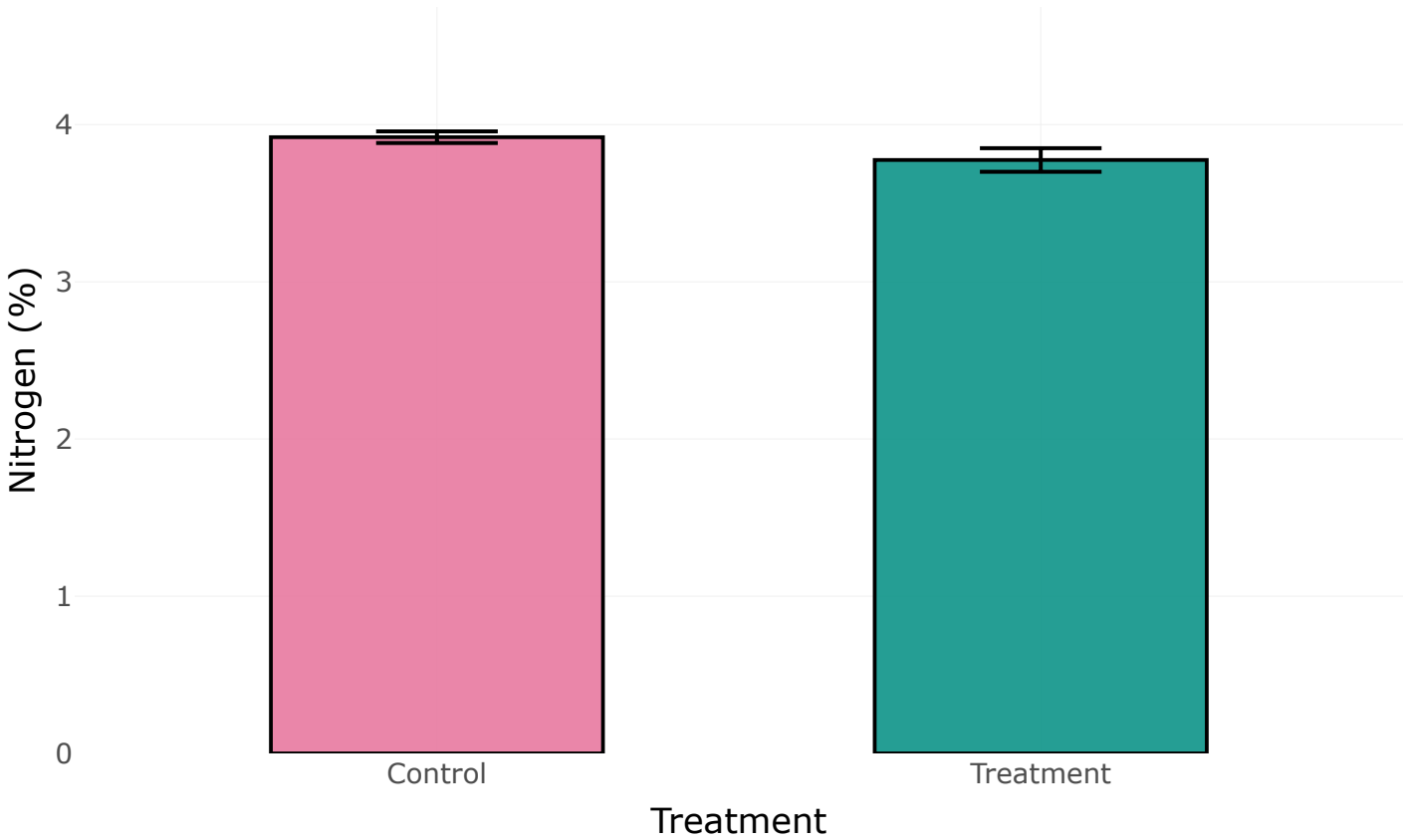
1.24

n = 5 Control · n = 4 Treatment

Bar Chart (%N)

Raw Data (%N)

Nitrogen in Corn Biomass (Mean %N ± SE)



C:N Ratio

The C:N ratio flips the perspective: a lower number means the tissue is more nitrogen-rich relative to its carbon content. If the Treatment had incorporated more nitrogen from the legume sod early on, we'd expect a lower C:N ratio in those plants. The values across both treatments were fairly close, which is consistent with the %N results, both systems delivered similar amounts of nitrogen to the plant at this growth stage.

TREATMENT — MEAN C:N RATIO

12.17 ↑ +6% vs Control

Control mean: 11.48

COHEN'S D (EFFECT SIZE)

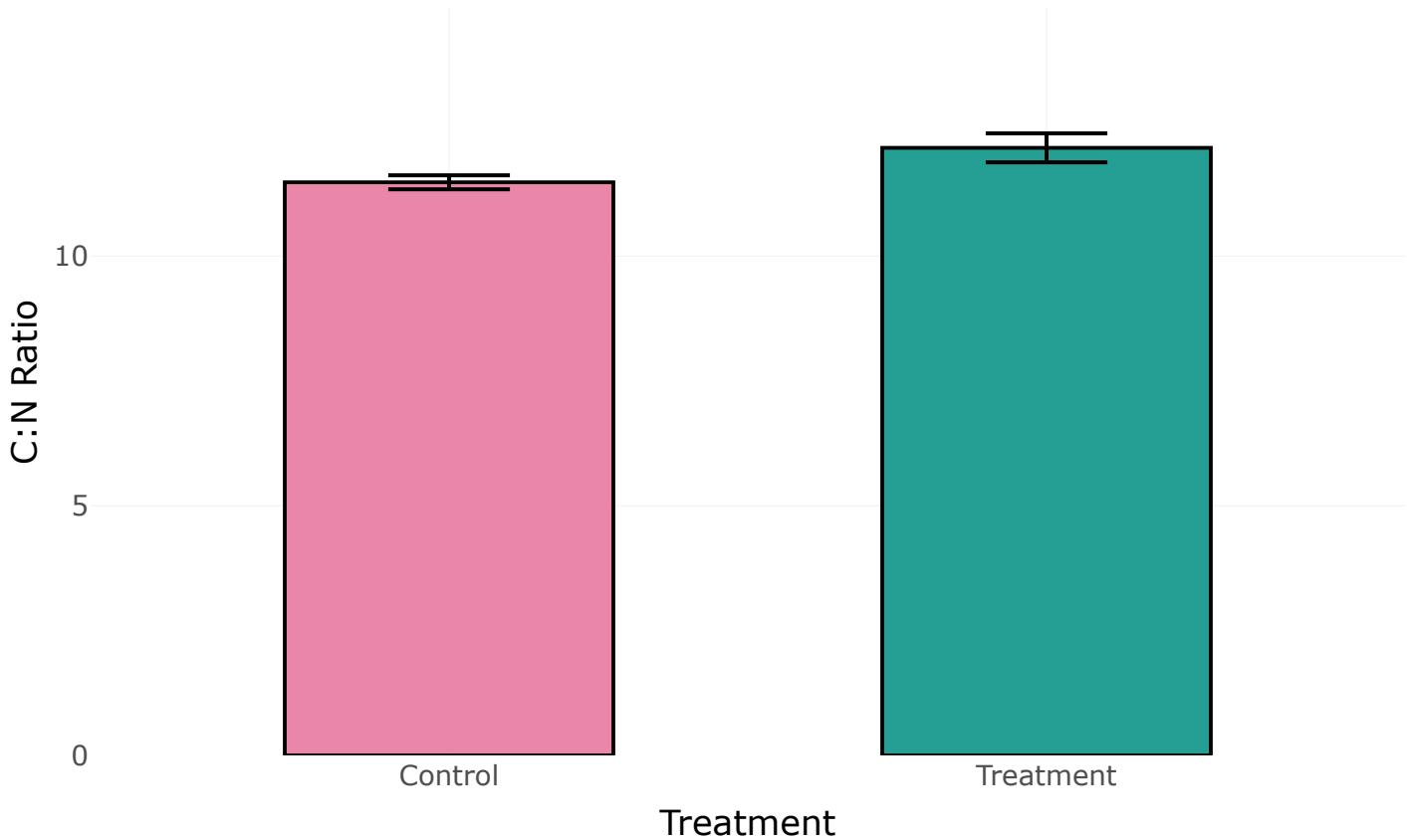
1.55

Lower C:N = more nitrogen-rich tissue

Bar Chart (C:N)

Raw Data (C:N)

C:N Ratio in Corn Biomass (Mean ± SE)



Cornstalk Nitrate Test (CSNT)

The Cornstalk Nitrate Test (CSNT) measures nitrate concentration in the lower cornstalk at the end of the season. It is a post-harvest diagnostic tool: values **below 250 ppm indicate insufficient nitrogen**, while the **optimal range is between 750 and 2000 ppm**. Because we only had one sample per treatment, this is a field-level observation rather than a statistical comparison.

CONTROL — CSNT

661 ppm

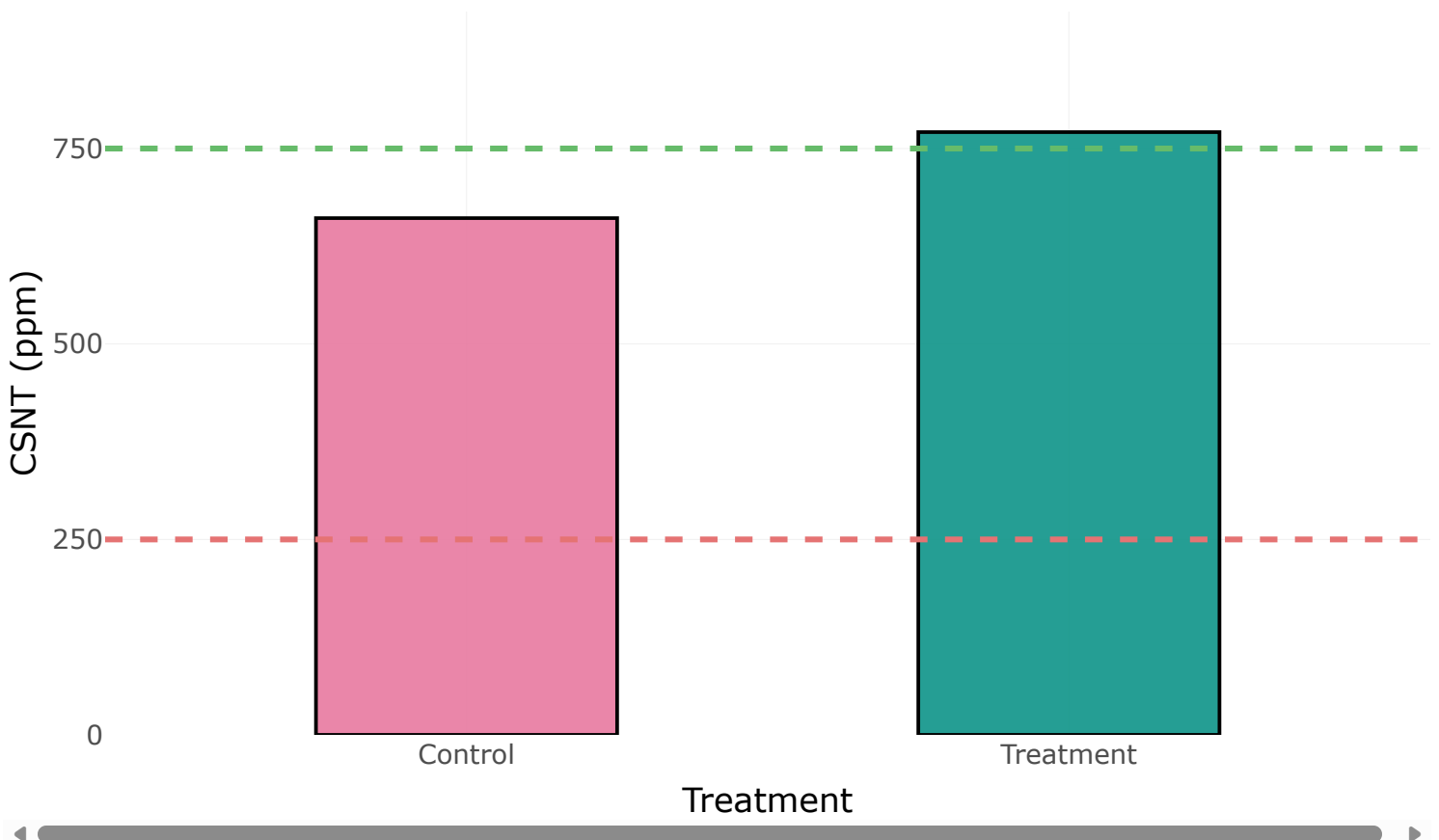
Threshold for sufficient N: 250 ppm

TREATMENT — CSNT

771 ppm

n = 1 per treatment — descriptive only

Cornstalk Nitrate Test (CSNT)



Soil health was assessed using the **Cornell Soil Health Test**. The overall quality score was **76/100 (High)**. Most indicators came back in the High to Very High range, which reflects years of good management. The areas to keep an eye on are surface and subsurface hardness (both Medium), which may indicate some compaction developing in the field, and active carbon (Medium), which reflects the level of readily available food for soil microbes. These are manageable, and continued cover cropping or diverse rotations can help move those numbers up over time.

Comprehensive Assessment of Soil Health

From the Cornell Soil Health Laboratory, Department of Soil and Crop Sciences
School of Integrative Plant Science, Cornell University, Ithaca, NY 14853
<https://soilhealthlab.cals.cornell.edu>



Grower: Rou Porter

Agricultural Service Provider:
Alex Yore
ay327@cornell.edu

Sample ID: QCC95
Field ID: OFE 2025 N AFTER CC PORTER G-7
Date Sampled: 06/30/2025
Coordinates: Latitude: 42.56611
Longitude: -76.720505

Measured Soil Textural Class: **loam**

Sand: **33%** - Silt: **48%** - Clay: **18%**

Group	Indicator	Value	Rating	Constraints
physical	<u>Predicted</u> Available Water Capacity	0.24	90	
physical	Surface Hardness	158	53	
physical	Subsurface Hardness	298	51	
physical	Aggregate Stability	57.3	93	
biological	Organic Matter Soil Organic Carbon: 2.7 / Total Carbon: 2.75 / Total Nitrogen: 0.28	4.7	97	
biological	<u>Predicted</u> Soil Protein	7.74	65	
biological	Soil Respiration	0.7	62	
biological	Active Carbon	511	52	
chemical	Soil pH	6.4	100	
chemical	Extractable Phosphorus	2.6	75	
chemical	Extractable Potassium	53.7	79	
chemical	Additional Nutrients Ca: 1831.1 / Mg: 131.2 / S: 5.5 Al: 16.3 / B: 0.2 / Cu: 0.07 Fe: 1.7 / Mn: 4.7 / Zn: 0.3		100	

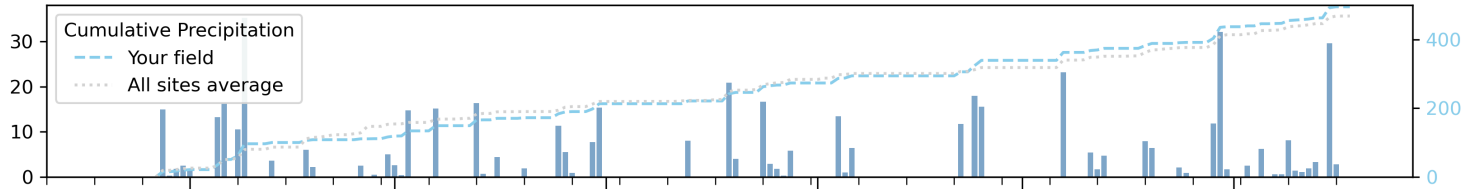
Overall Quality Score: **76 / High**

 [Open Full Soil Health Report \(PDF\)](#)

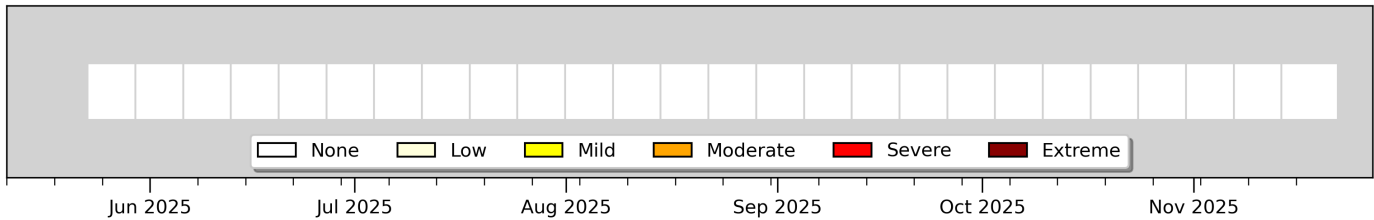
Weather & Drought Conditions

Cumulative precipitation at your field compared to the all-site average, alongside the weekly drought index from June through November 2025. Your field had a very consistent and well-distributed rainfall season in 2025. Precipitation tracked closely with the all-site average from June through October, finishing near 490 mm (about 19 inches) cumulative, slightly above the all-site average. There were no major drought episodes: the drought index stayed at None for every week shown. Moisture was not a limiting factor for your crop this year, which means the differences we see between Treatment and Control strips are more likely driven by management and soil factors than by weather.

Local vs All Site Average Precipitation 2025 (in mm)



Weekly Drought Index



Conclusions

LEAF NITROGEN STATUS (NBI)

NBI was nearly identical between the two systems — Treatment at **59.9** vs Control at **59.5**, a difference of less than 1%. Plowing the sod earlier did not change how much nitrogen the plant had available mid-season, at least as far as the NBI reading could detect. Both strips were in a similar range.

CHLOROPHYLL INDEX (CHL)

Chlorophyll was **9.1% higher** in the Treatment strips, a statistically significant difference. The leaves in the earlier-plowed strips were noticeably greener, suggesting a stronger early-season start. However, this did not translate into a higher NBI or better tissue nitrogen.

BIOMASS NITROGEN (%N)

Lab analysis showed Treatment plants had slightly **less nitrogen in their tissue** than Control (3.775% vs 3.92%, a difference of -3.7%). The effect size is small in absolute terms, and the sample size is too limited for a firm conclusion, but the direction is worth noting, as it runs counter to what the chlorophyll result suggested.

END-OF-SEASON CSNT

The stalk nitrate test showed Control at **661 ppm**, just below the 750 ppm optimal threshold, and Treatment at **771 ppm**, just inside the optimal range. Both crops finished the season reasonably well-supplied with nitrogen. With one sample per treatment, this is an observation to carry forward into the analysis of next season.

CORN YIELD

Treatment strips yielded 131.6 bu/ac vs 123.8 bu/ac in Control (+6.3%). With thousands of yield monitor points even small differences come out statistically significant, so the size of the difference (7.8 bu/ac) is the more meaningful number. Keep in mind that the Control area includes the field edges, which often yield less, while the Treatment strip runs through the middle of the field.